

Class Test 1: 28/07/2026

Write down the general solution of the following matrix given in REF form:

$A\vec{x} = \vec{b}$

$$\left[\begin{array}{ccccc|c} 2 & 0 & 0 & 5 & 2 & 4 \\ 0 & 0 & 1 & 3 & 7 & 6 \\ 0 & 0 & 0 & -1 & 1 & 8 \end{array} \right]$$

Also find two solutions of $A\vec{x} = \vec{0}$ where they are not scalar multiples of each other.*Proof.* The matrix is already in REF form. So we simply solve from bottom and ultimately get the general solution as follows:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 22 - 3.5\alpha \\ \beta \\ 30 - 10\alpha \\ \alpha - 8 \\ \alpha \end{bmatrix} = \alpha \begin{bmatrix} -3.5 \\ 0 \\ -10 \\ 1 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 22 \\ 0 \\ 30 \\ -8 \\ 0 \end{bmatrix}, \alpha, \beta \in F$$

□

For second part, simply consider these two

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

or something else idc.

Class Test 2: hewwo!! $> \omega <$